KAIZEN MATHS

Lesson starter

Sine Rule, Cosine Rule and Triangle Area

Cosine Rule & Mixed

Name _____

S1

In triangle ABC, a = 9 cm, b = 7 cm, and included angle C = 55°. Find c to 2 d.p.

S2

In triangle ABC, a = 11 cm, b = 12 cm, and c = 11 cm. Find angle C to 1 d.p.

S3

In triangle ABC, a = 12 cm, b = 11 cm, and included angle C = 86°. Find c to 2 d.p.

S4

In triangle ABC, a = 10 cm, b = 9 cm, and c = 10 cm. Find angle C to 1 d.p.

KAIZEN MATHS

Lesson starter

Sine Rule, Cosine Rule and Triangle Area

Cosine Rule & Mixed

Name _____

S1

In triangle ABC, a = 9 cm, b = 7 cm, and included angle C = 55°. Find c to 2 d.p.

S2

In triangle ABC, a = 11 cm, b = 12 cm, and c = 11 cm. Find angle C to 1 d.p.

S3

In triangle ABC, a = 12 cm, b = 11 cm, and included angle C = 86°. Find c to 2 d.p.

S4

In triangle ABC, a = 10 cm, b = 9 cm, and c = 10 cm. Find angle C to 1 d.p.

ALG-2B2B-L2 · S

Show your working

KAIZEN MATHS

Exit ticket

Sine Rule, Cosine Rule and Triangle Area

Cosine Rule & Mixed

Name _____

E1

In triangle ABC, a = 10 cm, b = 10 cm, and included angle C = 73°. Find c to 2 d.p.

E2

In triangle ABC, a = 8 cm, b = 7 cm, and c = 8 cm. Find angle C to 1 d.p.

E3

In triangle ABC, a = 12 cm, b = 9 cm, and included angle C = 122°. Find c to 2 d.p.

KAIZEN MATHS

Exit ticket

Sine Rule, Cosine Rule and Triangle Area

Cosine Rule & Mixed

Name _____

E1

In triangle ABC, a = 10 cm, b = 10 cm, and included angle C = 73°. Find c to 2 d.p.

E2

In triangle ABC, a = 8 cm, b = 7 cm, and c = 8 cm. Find angle C to 1 d.p.

E3

In triangle ABC, a = 12 cm, b = 9 cm, and included angle C = 122°. Find c to 2 d.p.

ALG-2B2B-L2 · E

Show your working

KAIZEN MATHS

Exit ticket

Sine Rule, Cosine Rule and Triangle Area

Cosine Rule & Mixed

Name _____

E1

In triangle ABC, a = 10 cm, b = 10 cm, and included angle C = 73°. Find c to 2 d.p.

E2

In triangle ABC, a = 8 cm, b = 7 cm, and c = 8 cm. Find angle C to 1 d.p.

E3

In triangle ABC, a = 12 cm, b = 9 cm, and included angle C = 122°. Find c to 2 d.p.

ALG-2B2B-L2 · E

Show your working

KAIZEN MATHS

Exit ticket

Sine Rule, Cosine Rule and Triangle Area

Cosine Rule & Mixed

Name _____

E1

In triangle ABC, a = 10 cm, b = 10 cm, and included angle C = 73°. Find c to 2 d.p.

E2

In triangle ABC, a = 8 cm, b = 7 cm, and c = 8 cm. Find angle C to 1 d.p.

E3

In triangle ABC, a = 12 cm, b = 9 cm, and included angle C = 122°. Find c to 2 d.p.

ALG-2B2B-L2 · E

Show your working

Teacher answers & working

Sine Rule, Cosine Rule and Triangle Area · Cosine Rule & Mixed

ALG-2B2B-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

In triangle ABC, a = 9 cm, b = 7 cm, and included angle C = 55°. Find c to 2 d.p.

Answer:

$c = 7.60$ cm

Working:

Cosine Rule: $c^2 = a^2 + b^2 - 2ab \cos C$

$c^2 = 9^2 + 7^2 - 2 \times 9 \times 7 \times \cos 55°$

$\cos 55° \approx 0.5736$

$c^2 \approx 57.7294$

$c \approx \sqrt{57.7294} \approx 7.5980$

$c = 7.60$ cm

S2

In triangle ABC, a = 11 cm, b = 12 cm, and c = 11 cm. Find angle C to 1 d.p.

Answer:

$C = 56.9°$

Working:

$\cos C = (a^2 + b^2 - c^2) / (2ab)$

$\cos C = (121 + 144 - 121) / (2 \times 11 \times 12)$

$\cos C \approx 0.5455$

$C \approx \cos^{-1}(0.5455) \approx 56.9°$

$C = 56.9°$

S3

In triangle ABC, a = 12 cm, b = 11 cm, and included angle C = 86°. Find c to 2 d.p.

Answer:

$c = 15.70$ cm

Working:

Cosine Rule: $c^2 = a^2 + b^2 - 2ab \cos C$

$c^2 = 12^2 + 11^2 - 2 \times 12 \times 11 \times \cos 86°$

$\cos 86° \approx 0.0698$

$c^2 \approx 246.5843$

$c \approx \sqrt{246.5843} \approx 15.7030$

$c = 15.70$ cm

Teacher answers & working

Sine Rule, Cosine Rule and Triangle Area · Cosine Rule & Mixed

ALG-2B2B-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S4

In triangle ABC, a = 10 cm, b = 9 cm, and c = 10 cm. Find angle C to 1 d.p.

Answer:

$C = 63.3°$

Working:

$\cos C = (a^2 + b^2 - c^2) / (2ab)$

$\cos C = (100 + 81 - 100) / (2 \times 10 \times 9)$

$\cos C \approx 0.4500$

$C \approx \cos^{-1}(0.4500) \approx 63.3°$

$C = 63.3°$

E1

In triangle ABC, a = 10 cm, b = 10 cm, and included angle C = 73°. Find c to 2 d.p.

Answer:

$c = 11.90$ cm

Working:

Cosine Rule: $c^2 = a^2 + b^2 - 2ab \cos C$

$c^2 = 10^2 + 10^2 - 2 \times 10 \times 10 \times \cos 73°$

$\cos 73° \approx 0.2924$

$c^2 \approx 141.5257$

$c \approx \sqrt{141.5257} \approx 11.8965$

$c = 11.90$ cm

E2

In triangle ABC, a = 8 cm, b = 7 cm, and c = 8 cm. Find angle C to 1 d.p.

Answer:

$C = 64.1°$

Working:

$\cos C = (a^2 + b^2 - c^2) / (2ab)$

$\cos C = (64 + 49 - 64) / (2 \times 8 \times 7)$

$\cos C \approx 0.4375$

$C \approx \cos^{-1}(0.4375) \approx 64.1°$

$C = 64.1°$

Teacher answers & working

Sine Rule, Cosine Rule and Triangle Area · Cosine Rule & Mixed

ALG-2B2B-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E3

In triangle ABC, a = 12 cm, b = 9 cm, and included angle C = 122°. Find c to 2 d.p.

Answer:

$c = 18.42$ cm

Working:

Cosine Rule: $c^2 = a^2 + b^2 - 2ab \cos C$

$c^2 = 12^2 + 9^2 - 2 \times 12 \times 9 \times \cos 122°$

$\cos 122° \approx -0.5299$

$c^2 \approx 339.4626$

$c \approx \sqrt{339.4626} \approx 18.4245$

$c = 18.42$ cm